SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I G. Yashmita (192511032) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Yashmita

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1-AT3 - CSA1520 - CLOUD COMPUTING

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7/10/26

Short Answer Test

1. Define cloud computing & Explain any four characteristics that make it different from traditional on-premise computing

2. Trace the evolution of cloud computing from hardware evolution its internet software evolution & show how these changes enabled modern cloud platforms.

3. Explain the role of server virtualization in Cloud Computing. Differentiate between virtualization as an enabling technology & Cloud computing as a Service model.

4. Discuss the role of data centers in delivering cloud services.

5. Differentiate IaaS, PaaS, SaaS, & XaaS with 1 suitable example of real world application.

1. Cloud computing

Cloud computing is a technology that provides computing resources such as memory, storage and network access. We can work on that server wherever we needed, it's like Pay-When-you-use.

Characteristics

1. On-demand service :

It can be used whenever we needed it. The service provider doesn't have to do anything. We can use it by paying.

2. Broad Network access :

It is that the server can be connected using any laptop, computers, ... Since, the network is connected through internet. So, anyone can access it.

3. Resource pooling

It is That, server can be connected using multiple users in which the resources are connected.

4. High availability

It is highly available through internet, since we can connect it anytime.

Conclusion :

In the traditional cloud server, that service provider has to do all the works. But, in present we can use it anytime as pay use.

2. Evolution of Cloud Computing in data centers.

* Networking and internet

The development of computer networks & internet allowed computers to communicate and share resources to different location.

* Client Server Computing

Applications began to use separate client and server system.

* Virtualization

It allows multiple virtual machines in single physical server.

* Internet :

Cloud provides combined virtualization high speed networks, data centers to provide IaaS, PaaS, SaaS & XaaS

Conclusion :

The above evolution made changes in modern Cloud platforms.

3. Server virtualization

In Server virtualization, we can connect multiple virtual machine to one single physical layer of server using software called hypervisors.

Role in Cloud computing:

Improves utilization of physical hardware.

Provides Isolation between different users.

It makes it easier to create more and manage servers.

It mainly Reduces infrastructure and maintanence cost.

Virtualization	Cloud computing
* Enabling technology	Service delivery model
* Creates mainly focus on Efficient resources.	Focus on delivering service to users
* Create virtual resource from physical resource.	provides computing resource through internet
* Ex: physical server virtual machines	Ex: User -> Internet -> AWS -> virtual server

4. Role of Data centers:

A data center is facility that will contain more no. of servers, storage systems, memory.

1. Computing resources:

Data centers contains physical server that provide processing power.

2. Networking:

High speed networking equipment that connects the services.

3. Data storage

It will store workfile, applications and Backup also.

4. Scalability

Cloud provides servers & storage when demand increases.

5. Reliability

It helps to keep cloud services reliable to multiple network.

5.

Iaas

* Infrastructure as a service
* This Iaas provides basic computing infrastructure such as storage networking servers

Ex: AWS

Paas: * Platform as a Service
* Provides basic computing which makes an application

Ex: Google doc

Saas:

* Software as a Service
* provides software based applications.

Ex: Google drive